

AI Glasses RK3576 Chip-down BOM V2.3

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Date

2026-07-01

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U2 / C002 - RK806S-5

U3 / C003 - LPDDR4X 4GB (MPN TBD)

U4 / C004 - eMMC 5.1 32GB (MPN TBD)

Y1 / C005 - 24 MHz 10 ppm XTAL

U5 / C006 - MX25U6432F

U6 / C007 - TPS61088

L1 / C008 - Boost inductor (TBD)

U11 / C015 - FCU760KAAMD

U12 / C016 - TPS62825

L2 / C016c - Wi-Fi buck inductor (TBD)

U13 / C016b - TPS22917DBVR

C1 / C015-C - 47uF 6.3V X5R

RT3 / C043 - 10k NTC B=3435

U7 / C009 - nRF54L15-QFN48

U8 / C010 - nPM1300

U9 / C011 - NDP120

U10 / C012 - BMI270

J6 / C018 - BLE 2.4 GHz FPC/PCB ant

J7 / C017 - Wi-Fi 2.4/5 GHz FPC ant

U14 / C019 - IMX415-AAQR module (custom FPC)

U15 / C020 - TPS62840
L3 / C020b - Cam 1V1 buck inductor (TBD)
U16 / C021 - TLV75529PDRVR
U17 / C022 - TPS22917DBVR
U18 / C023a - TPD4E05U06
U19 / C023b - TPD4E05U06
MK1 / C013a - T5837 (wake mic)
MK2 / C013b - T5837 (array mic 1)
MK3 / C013c - T5837 (array mic 2)
MK4 / C014 - T5837 (4th mic)
U20 / C024 - MAX98360A
U21 / C024b - TPS22917DBVR
LS1 / C025 - 8 ohm 0.5-1 W speaker
LS2 / C026 - 2nd speaker pad
U22 / C027 - DRV2605L
M1 / C028 - LRA/ERM motor
BT1 / C029 - LP451165 300mAh (R)
BT2 / C030 - LP451165 300mAh (L)
F1 / C031a - PTC/fuse (R branch)
F2 / C031b - PTC/fuse (L branch)
RS2 / C044-RCL - 10 mohm 1%
RS3 / C044-RCR - 10 mohm 1%
RS1 / C044-RBAT - 10 mohm 1% 1W
R21 / C032-BYP - 0R (pack-PCM baseline)
U23 / C032 - BQ2970
Q1 / C032-Q - Dual N-MOSFET (b2b)
R22 / C032-Rv - 330R
C2 / C032-Cv - 0.1uF
R23 / C032-Rm - 2k
RT1 / C033a - 10k NTC (R cell)
RT2 / C033b - 10k NTC (L cell)
TP2 / C033-TP - Battery test points
J1 / C034 - Magnetic pogo 4-6p
J2 / C035 - USB-C 16p USB2-only
U24 / C036 - TPD2E001
R1 / C037a - 5.1k 1%
R2 / C037b - 5.1k 1%
J3 / C038 - Hirose FH26W 0.3mm FPC
J4 / C039 - Custom hinge FPC 6-10mm
J5 / C040 - U.FL / I-PEX MHF
TP1 / C041 - UART/SWD pogo pads
SW1 / C042a - Power/PWRKEY

SW2 / C042b - Reset
SW3 / C042c - MaskROM/Recovery
RS4 / C044-RSOC - 10 mohm 1%
RS5 / C044-RWIFI - 50 mohm 1%
RS6 / C044-RAUD - 50 mohm 1%
RS7 / C044-RC1 - 100 mohm 1%
RS8 / C044-RC8 - 100 mohm 1%
RS9 / C044-RC29 - 100 mohm 1%
U25 / C044a - INA238 (I_BAT_TOTAL)
U26 / C044-ICL - INA238 (I_CELL_L)
U27 / C044-ICR - INA238 (I_CELL_R)
U28 / C044c - INA238 (I_SOC_5V)
U29 / C044e - INA238 (I_WIFI)
U30 / C044f - INA238 (I_AUDIO)
U31 / C044-IC1 - INA238 (I_CAM_1V1)
U32 / C044-IC8 - INA238 (I_CAM_1V8)
U33 / C044-IC29 - INA238 (I_CAM_2V9)
R11 / C046a - 100k
R12 / C046b - 100k
R13 / C046c - 100k
R14 / C046d - 100k
R15 / C046e - 100k
R16 / C046f - 100k
R17 / C046g - 100k
R18 / C046h - 100k
R19 / C046i - 100k
R20 / C046j - 100k
R3 / C045a - 2.2k 1%
R4 / C045b - 2.2k 1%
R5 / C045c - 2.2k 1%
R6 / C045d - 2.2k 1%
R9 / C045e - 2.2k 1%
R10 / C045f - 2.2k 1%

Current schematic-aligned BOM, power, and layout-gate report

- Generated: 2026-07-01
- Source of truth: [v2_chipdown/scripts/chipdown_bom.py](#) and [v2_chipdown/config/power_budget_v2.yaml](#)
- Schematic: [v2_chipdown/hardware/ai_glasses_v2_chipdown.kicad_sch](#)
- Exported BOM/netlist/PDF: [v2_chipdown/reports/*](#)
- This report is schematic-aligned. If an older PDF disagrees with this document, use this document.

1. Read this first

Item	Current value
Schematic scope	Pre-PCB-layout functional-block schematic
Components / nets	96 components / 127 nets
Assembly states	DNP=19, Fit=58, HOLD=12, TBD=7
Board split	R-Temple Compute Board=31, L-Temple AON/Power Board=15, Temple Rears (batt/spkr/ant)=19, Front Sensor Board=23, EVT Debug Tail=8
ERC	Report includes: Errors, Warnings, Exclusions; ** ERC messages: 0 Errors 0 Warnings 0
PCB layout release	NOT released; Gate 0/G00F/G05/G07/G08/G10/G12 still block layout
Phase 1.5	Required mechanical/electrical floorplan before formal PCB layout
First EVT hinge stance	Fixed-temple EVT is allowed; J4/C039 is an interconnect/FPC placeholder, not a frozen mechanical hinge

Status legend: Fit = populate first build; DNP = land only; HOLD = candidate blocked by a gate; TBD = MPN/spec undecided. Power numbers are project budget estimates until EVT shunt measurements replace them.

2. System power boundary

Power state	Current load (mW)	Target band (mW)	Battery current @3.7 V	Runtime on LP451165 x2 1S2P
Deep Off (AON only, RK3576 down)	22	20-50	6 mA	91 h

Power state	Current load (mW)	Target band (mW)	Battery current @3.7 V	Runtime on LP451165 x2 1S2P
Quick Standby (DDR retention / light sleep)	134	80-200	36 mA	15 h
Phone-assisted safety ID (blended)	362	250-450	98 mA	5.5 h (331 min)
Mixed motion use (blended)	450	<=500-800	122 mA	4.4 h (266 min)
Record Active (1080p, ISP+encode)	1290	1000-1500	349 mA	1.5 h (93 min)
AI Burst (NPU + encode + Wi-Fi)	2065	1500-2500	558 mA	1.0 h (58 min)

Recommended pack: LP451165 x2, 1S2P, 600 mAh / 2.22 Wh nominal, one cell per temple. Baseline current model uses 1290 mW for continuous 1080p record and 2065 mW for AI burst.

3. Subsystem power model

Subsystem	Domain	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record
BLE MCU (nRF-class, always-on)	AON	2	3	8	6	5	8
Audio DSP / wake-word (always listening)	AON	12	12	15	15	15	20
IMU BMI270 (low-power)	AON	1	1	1	1	1	1
MEMS mic array (wake mic always; array when active)	AON	3	3	6	6	8	10
Fuel gauge + NTC + protection	AON	1	1	1	1	1	1
AON Buck quiescent + regulation loss	AON	3	4	6	6	10	15

Subsystem	Domain	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record
RK3576 SoC + LPDDR4X + eMMC + PMIC	COMPUTE	0	90	150	250	850	1500
Custom 1080p camera (sensor + ISP share)	CAMERA	0	0	0	30	250	300
Wi-Fi (on-demand)	RADIO	0	0	150	100	80	120
Audio amp + speaker (avg)	AUDIO	0	0	15	15	30	30
Compute-isl and load switches / PMIC loss	COMPUTE	0	20	10	20	40	60
TOTAL		**22**	**134**	**362**	**450**	**1290**	**2065**

4. What changed from the attached PDF

- The Wi-Fi module is the current schematic part **U11 / C015 = FCU760KAAMD**, status HOLD.
- The IMU is **U10 / C012 = BMI270**, status Fit.
- The current baseline power totals are 22 / 134 / 362 / 450 / 1290 / 2065 mW.
- The current BOM is not only C001-C045; it is 96 schematic components with expanded shunts, INA238 monitors, passives, load switches, and C046 default-off pull-downs.
- C044 is expanded into RS1-RS9 plus U25-U33 current monitors; C045/C046 passives are explicit pre-layout controls.
- PCB layout is still not released; unresolved mechanical/battery/FPC/RF/camera/AON gates remain.
- A new Phase 1.5 floorplan is required between ERC-clean schematic and formal PCB layout.

5. Board and mechanical partition

Region	Current schematic contents	EVT note
Front Sensor Board	IMX415 custom FPC module, 3 fitted T5837 mics, 4th mic DNP, camera rails, MIPI ESD, front FPC	Final module pinout/lens/lane count remains G10 HOLD

Region	Current schematic contents	EVT note
R-Temple Compute Board	RK3576, RK806S, LPDDR4X, eMMC, TPS61088, FCU760K Wi-Fi, Wi-Fi buck/load switch, MAX98360A audio	Heat and boost droop gates remain open
L-Temple AON/Power Board	nRF54L15, nPM1300, NDP120, BMI270, optional protection, main shunt/INA	AON average ≤ 25 mW and ceiling ≤ 50 mW must be measured
Temple Rears	LP451165 x2, branch fuses/shunts, speaker, antennas, pogo, hinge/FPC interconnect placeholder	First EVT can be fixed-temple; reserve future FPC/hinge corridor
EVT Debug Tail	USB-C, ESD, CC resistors, SWD/UART pads, power/reset/recovery buttons	Bring-up only; production can hide/remove

5.5. Phase 1.5 mechanical/electrical floorplan

Before formal PCB placement/routing, create a no-route fit floorplan. This is a CAD + EE deliverable, not a routed board.

Required output	Contents
Right-temple top/side view	usable length/width/height; Compute PCB; RK3576/LPDDR/RK806/eMMC/Wi-Fi/boost-inductor zones; battery; speaker; antenna keep-out; pogo; FPC corridor; shell/foam/swell allowance
Left-temple top/side view	usable length/width/height; AON/power PCB; nRF/nPM/NDP/BMI270 zones; battery; BLE antenna keep-out; pogo/FPC corridor; shell/foam/swell allowance
KiCad no-route floorplan	board frames plus RK3576, LPDDR, RK806, eMMC, Wi-Fi, boost inductor, FPC connectors, battery 3D/envelope blocks, speaker and antenna keep-outs
Pass/fail decision	prove the current schematic architecture fits before routing; if not, change battery/partition/connectors/DNP features/local temple width

Formal routing starts only after this floorplan passes. If it fails, do not route; change the architecture first.

6. Current BOM by board

R-Temple Compute Board

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U1	C001	RK3576	Fit	P0	AI_Glasses_V2:VERIFY_FC CSP698L_16x17	G01 (RK3576 datasheet/HDG/ballmap/DDR guide)
U2	C002	RK806S-5	HOLD	P0	AI_Glasses_V2:VERIFY_QFN	G03 (RK806S MPN/inductors/caps/timing per Radxa/Rockchip FAE)
U3	C003	LPDDR4X 4GB (MPN TBD)	HOLD	P0	AI_Glasses_V2:VERIFY_BGA	G02 (LPDDR4X MPN/topology/placement, DDR review + length report)
U4	C004	eMMC 5.1 32GB (MPN TBD)	HOLD	P0	AI_Glasses_V2:VERIFY_BGA153	G04 (eMMC MPN + BSP/bootloader; cold-boot + power-loss recovery)
Y1	C005	24 MHz 10 ppm XTAL	Fit	P0	AI_Glasses_V2:Crystal_SMD_3225	G01 (reuse reference CL/ESR + placement distance per HDG)
U5	C006	MX25U6432F	DNP	P1	AI_Glasses_V2:USON-8	Boot flow decides whether production needs it
U6	C007	TPS61088	HOLD	P0	AI_Glasses_V2:VQFN-22_4.5x3.5	G06/G09 (measured RK3576 boot+AI peak, soft-start, droop, thermal)
L1	C008	Boost inductor (TBD)	TBD	P0	AI_Glasses_V2:VERIFY_L_5x5	G06/G13 (Isat covers boot+AI peak; height <=2mm; thermal)

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U11	C015	FCU760KAAMD	HOLD	P0	AI_Glasses_V2:VERIFY_LCC_13x12.2_Quetel	G05 (Quectel HW Design + RK3576 BSP driver/firmware/enum) + official LCC land pattern
U12	C016	TPS62825	Fit	P0	AI_Glasses_V2:QFN_1.5x1.5	G05/G06 (inductor/caps per module 353mA TX peak + ripple)
L2	C016c	Wi-Fi buck inductor (TBD)	Fit	P0	AI_Glasses_V2:VERIFY_L_2x2	Per TPS62825 peak+ripple
U13	C016b	TPS22917DBVR	Fit	P0	Package_TO_SOT_SMD:SO T-23-5	G05 (turn-on sequence w/ module)
C1	C015-C	47uF 6.3V X5R	Fit	P0	AI_Glasses_V2:C_0805	Local bulk per Quectel HW Design
RT3	C043	10k NTC B=3435	Fit	P0	AI_Glasses_V2:R_0402	Probe position per thermal sim
U20	C024	MAX98360A	Fit	P0	AI_Glasses_V2:VERIFY_WLP_QFN	Verify Z/cavity/EMI/peak power
U21	C024b	TPS22917DBVR	Fit	P0	Package_TO_SOT_SMD:SO T-23-5	Amp must be fully power-downable (V1 lesson)
RS4	C044-RSOC	10 mohm 1%	Fit	P0	AI_Glasses_V2:R_0805_shunt	5-20 mohm (2A x 10m=20mV; avoid boost UVLO)
RS5	C044-RWIFI	50 mohm 1%	Fit	P0	AI_Glasses_V2:R_0603_shunt	20-100 mohm
RS6	C044-RAUD	50 mohm 1%	Fit	P0	AI_Glasses_V2:R_0603_shunt	20-100 mohm
U28	C044c	INA238 (I_SOC_5V)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U29	C044e	INA238 (I_WIFI)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only
U30	C044f	INA238 (I_AUDIO)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only
R11	C046a	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R12	C046b	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R16	C046f	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R17	C046g	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R18	C046h	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4/section 9: FCU760K CHIP_EN default low)
R19	C046i	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R20	C046j	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R9	C045e	2.2k 1%	Fit	P0	AI_Glasses_V2:R_0402	Confirm vs SoC PMIC bus
R10	C045f	2.2k 1%	Fit	P0	AI_Glasses_V2:R_0402	Confirm vs SoC PMIC bus

L-Temple AON/Power Board

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U7	C009	nRF54L15-QF N48	Fit	P0	AI_Glasses_V2:QFN-48_6x6	EVT-frozen (RF layout/SD K/package to freeze early)
U8	C010	nPM1300	HOLD	P0	AI_Glasses_V2:QFN_5x5	G08 (AON <=25/50mW) - configure via nPM PowerUP on EK before board

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U9	C011	NDP120	HOLD	P0	AI_Glasses_V2:VERIFY_QFN_5x5	Full datasheet + dev kit + measured listening power + NDA/licensing
U10	C012	BMI270	Fit	P0	Package_LGA: LGA-14_2.5x3mm_P0.5mm	Evaluate false-trigger under real frame vibration
U22	C027	DRV2605L	DNP	P1	Package_SO: MSOP-10_3x3mm_P0.5mm	Vibration P1 - may DNP on first board (IMU/mic coupling)
RS1	C044-RBAT	10 mohm 1% 1W	Fit	P0	AI_Glasses_V2:R_1206_shunt	I_BAT_TOTAL; the one production-kept sense path
R21	C032-BYP	0R (pack-PCM baseline)	Fit	P0	AI_Glasses_V2:R_0805	Populate 0R OR discrete protection, never both (section 17)
U23	C032	BQ2970	DNP	P0	Package_TO_SOT_SMD:SO T-23-5	Keep ONE protection scheme; confirm suffix OVP/UVLP/OC P vs nPM1300/boost UVLO (section 17)
Q1	C032-Q	Dual N-MOSFET (b2b)	DNP	P0	AI_Glasses_V2:VERIFY_DF N_2FET	Vds/RDSon/Vgs/peak per BQ2970 app (section 17)
R22	C032-Rv	330R	DNP	P0	AI_Glasses_V2:R_0402	Per BQ2970 app
C2	C032-Cv	0.1uF	DNP	P0	AI_Glasses_V2:C_0402	Per BQ2970 app
R23	C032-Rm	2k	DNP	P0	AI_Glasses_V2:R_0402	Per BQ2970 app (V- sense)
U25	C044a	INA238 (I_BAT_TOTAL)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A; production may keep this one

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
R3	C045a	2.2k 1%	Fit	P0	AI_Glasses_V2:R_0402	Confirm vs total AON bus capacitance
R4	C045b	2.2k 1%	Fit	P0	AI_Glasses_V2:R_0402	Confirm vs total AON bus capacitance

Temple Rears (batt/spkr/ant)

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
J6	C018	BLE 2.4 GHz FPC/PCB ant	TBD	P0	AI_Glasses_V2:VERIFY_ANT	Worn-state tuning; clear of battery/metal
J7	C017	Wi-Fi 2.4/5 GHz FPC ant	TBD	P0	AI_Glasses_V2:VERIFY_ANT	Head-loaded RSSI/throughput /SAR; keep-out
LS1	C025	8 ohm 0.5-1 W speaker	TBD	P0	AI_Glasses_V2:VERIFY_SPK	MPN after acoustic cavity/ volume/leak test
LS2	C026	2nd speaker pad	DNP	P1	AI_Glasses_V2:VERIFY_SPK	Decide dual-speaker at EVT-B
M1	C028	LRA/ERM motor	DNP	P1	AI_Glasses_V2:VERIFY_LRA	Vibration P1
BT1	C029	LP451165 300mAh (R)	HOLD	P0	AI_Glasses_V2:VERIFY_CEL_L_4.5x11x65	G07 (full datasheet, >=2C, IR, cycles, cert)
BT2	C030	LP451165 300mAh (L)	HOLD	P0	AI_Glasses_V2:VERIFY_CEL_L_70x12.8x5.6_maxenv	G07
F1	C031a	PTC/fuse (R branch)	TBD	P0	AI_Glasses_V2:VERIFY_FUSE	Rating from peak-current calc (> branch peak, < FPC rating)
F2	C031b	PTC/fuse (L branch)	TBD	P0	AI_Glasses_V2:VERIFY_FUSE	Rating from peak-current calc

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
RS2	C044-RCL	10 mohm 1%	Fit	P0	AI_Glasses_V2:R_0805_shunt	Branch current-share (section 7/section 29)
RS3	C044-RCR	10 mohm 1%	Fit	P0	AI_Glasses_V2:R_0805_shunt	Branch current-share (section 7/section 29)
RT1	C033a	10k NTC (R cell)	Fit	P0	AI_Glasses_V2:R_0402	NTC curve matched to nPM1300 charger config
RT2	C033b	10k NTC (L cell)	Fit	P0	AI_Glasses_V2:R_0402	NTC curve matched to nPM1300 charger config
TP2	C033-TP	Battery test points	Fit	P0	AI_Glasses_V2:TestPad_Set	Cell/pack voltage + NTC probe access (section 21)
J1	C034	Magnetic pogo 4-6p	TBD	P0	AI_Glasses_V2:VERIFY_POGO	Magnet direction, sweat corrosion, short, cycle life
J4	C039	Custom hinge FPC 6-10mm	HOLD	P0	AI_Glasses_V2:VERIFY_HINGE_FPC	G12 (impedance, bend radius, life, hinge interference)
J5	C040	U.FL / I-PEX MHF	Fit	P0	Connector_Coaxial:U.FL_Hirose_U.FL-R-SMT-1_Vertical	Connector height + mating life
U26	C044-ICL	INA238 (I_CELL_L)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only (current-share)
U27	C044-ICR	INA238 (I_CELL_R)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only (current-share)

Front Sensor Board

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U14	C019	IMX415-AAQR module (custom FPC)	HOLD	P0	AI_Glasses_V2:VERIFY_CAM_MODULE	G10 (module lens/FOV/FPC pinout/lane/supply/timing from vendor)
U15	C020	TPS62840	Fit	P0	AI_Glasses_V2:VERIFY_WSON	Output current/noise per final module (DVDD ~250mA max + margin)
L3	C020b	Cam 1V1 buck inductor (TBD)	Fit	P0	AI_Glasses_V2:VERIFY_L_1.6x0.8	Per TPS62840 design
U16	C021	TLV75529PDR VR	Fit	P0	AI_Glasses_V2:WSON-6_2x2	Verify vs IMX415 AVDD peak (156mA) + PSRR
U17	C022	TPS22917DBVR	Fit	P0	Package_TO_SOT_SMD:SO T-23-5	Check reverse block, ramp, QOD, logic level
U18	C023a	TPD4E05U06	Fit	P0	AI_Glasses_V2:USON	Low-cap array near FPC entry
U19	C023b	TPD4E05U06	Fit	P0	AI_Glasses_V2:USON	Low-cap array near FPC entry
MK1	C013a	T5837 (wake mic)	Fit	P0	AI_Glasses_V2:LGA-5_3.5x2.65	G11 (mic coords/ports/wind/wake + AEC/beamforming)
MK2	C013b	T5837 (array mic 1)	Fit	P0	AI_Glasses_V2:LGA-5_3.5x2.65	G11
MK3	C013c	T5837 (array mic 2)	Fit	P0	AI_Glasses_V2:LGA-5_3.5x2.65	G11
MK4	C014	T5837 (4th mic)	DNP	P1	AI_Glasses_V2:LGA-5_3.5x2.65	G11 (populate only if array sim/proto needs it)
J3	C038	Hirose FH26W 0.3mm FPC	HOLD	P0	AI_Glasses_V2:VERIFY_FPC_FH26W	G12 (pin count from final cam lane/mic/power split)

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
RS7	C044-RC1	100 mohm 1%	Fit	P0	AI_Glasses_V2:R_0603_shunt	Per-rail Kelvin (section 23/section 29); DVDD ~250mA
RS8	C044-RC8	100 mohm 1%	Fit	P0	AI_Glasses_V2:R_0402_shunt	Per-rail Kelvin (section 23/section 29); IOVDD ~3mA
RS9	C044-RC29	100 mohm 1%	Fit	P0	AI_Glasses_V2:R_0603_shunt	Per-rail Kelvin (section 23/section 29); AVDD ~156mA
U31	C044-IC1	INA238 (I_CAM_1V1)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only
U32	C044-IC8	INA238 (I_CAM_1V8)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only
U33	C044-IC29	INA238 (I_CAM_2V9)	DNP	P1	AI_Glasses_V2:VSSOP-10	Power Gate - EVT-A only
R13	C046c	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R14	C046d	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R15	C046e	100k	Fit	P0	AI_Glasses_V2:R_0402	Default OFF (section 4)
R5	C045c	2.2k 1%	Fit	P0	AI_Glasses_V2:R_0402	Confirm vs camera bus capacitance
R6	C045d	2.2k 1%	Fit	P0	AI_Glasses_V2:R_0402	Confirm vs camera bus capacitance

EVT Debug Tail

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
J2	C035	USB-C 16p USB2-only	Fit	P0	AI_Glasses_V2:USB_C_16P_MidMount	Connector height + shell opening co-freeze

Ref	BOM ID	Value	Status	Pri	Package / footprint placeholder	Gate
U24	C036	TPD2E001	Fit	P0	Package_TO_SOT_SMD:SO T-23-3	Keep ~90 ohm diff + continuous ref gnd
R1	C037a	5.1k 1%	Fit	P0	AI_Glasses_V2:R_0402	USB-C spec
R2	C037b	5.1k 1%	Fit	P0	AI_Glasses_V2:R_0402	USB-C spec
TP1	C041	UART/SWD pogo pads	Fit	P0	AI_Glasses_V2:TestPad_Set	Voltage domain clearly labelled (no 3.3V into 1.8V IO)
SW1	C042a	Power/PWRKEY	Fit	P0	Button_Switch_SMD:SW_Push_1P1T_NOCK_KMR2	First chip-down must recover from boot failure
SW2	C042b	Reset	Fit	P0	Button_Switch_SMD:SW_Push_1P1T_NOCK_KMR2	Bring-up
SW3	C042c	MaskROM/Recovery	Fit	P0	Button_Switch_SMD:SW_Push_1P1T_NOCK_KMR2	Bring-up (recover from bad boot)

7. Per-component power allocation

Values below are battery-side project-budget mW and sum exactly to the current subsystem totals. A zero means DNP, passive/source, or already included in its owning domain; it does not mean zero parasitic physics.

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U1	C001	RK3576	Fit	0	20	80	150	600	1150	RK3576 project allocation inside compute core row

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U2	C002	RK806S-5	HOLD	0	10	15	25	70	100	RK806S /PMIC rail-management allocation
U3	C003	LPDDR 4X 4GB (MPN TBD)	HOLD	0	55	45	60	130	180	LPDDR 4X retention/activity allocation
U4	C004	eMMC 5.1 32GB (MPN TBD)	HOLD	0	5	10	15	50	70	eMMC standby/write allocation
Y1	C005	24 MHz 10 ppm XTAL	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U5	C006	MX25U 6432F	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U6	C007	TPS61088	HOLD	0	20	10	20	40	60	compute-island boost/load-switch/PMIC loss row
L1	C008	Boost inductor (TBD)	TBD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U11	C015	FCU760 KAAMD	HOLD	0	0	145	95	75	115	FCU760 K on-demand Wi-Fi activity average
U12	C016	TPS62825	Fit	0	0	5	5	5	5	Wi-Fi buck conversion-loss all ocaation
L2	C016c	Wi-Fi buck inductor (TBD)	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U13	C016b	TPS22917DBV R	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
C1	C015-C	47uF 6.3V X5R	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RT3	C043	10k NTC B=3435	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
U7	C009	nRF54L 15-QFN 48	Fit	2	3	8	6	5	8	nRF-cla ss BLE MCU/cu rrent-sta te-mach ine average
U8	C010	nPM130 0	HOLD	3	4	6	6	10	15	nPM130 0 AON buck qui escent+ regulati on loss
U9	C011	NDP120	HOLD	12	12	15	15	15	20	NDP120 -class w ake-wor d/listeni ng budget
U10	C012	BMI270	Fit	1	1	1	1	1	1	BMI270 low-pow er motion budget
J6	C018	BLE 2.4 GHz FP C/PCB ant	TBD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
J7	C017	Wi-Fi 2.4/5 GHz FPC ant	TBD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U14	C019	IMX415- AAQR module (custom FPC)	HOLD	0	0	0	25	220	265	custom 1080p camera module baseline allocatio n

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U15	C020	TPS62840	Fit	0	0	0	2	15	18	camera 1.1 V buck loss allocation
L3	C020b	Cam 1V1 buck inductor (TBD)	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U16	C021	TLV755 29PDR VR	Fit	0	0	0	3	14	16	camera 2.9 V LDO loss allocation
U17	C022	TPS229 17DBV R	Fit	0	0	0	0	1	1	camera 1.8 V load-switch loss allocation
U18	C023a	TPD4E05U06	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U19	C023b	TPD4E05U06	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
MK1	C013a	T5837 (wake mic)	Fit	3	3	3	3	3	3	wake mic always-on share of MEMS mic row

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
MK2	C013b	T5837 (array mic 1)	Fit	0	0	1.5	1.5	2.5	3.5	array mic share of MEMS mic row
MK3	C013c	T5837 (array mic 2)	Fit	0	0	1.5	1.5	2.5	3.5	array mic share of MEMS mic row
MK4	C014	T5837 (4th mic)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U20	C024	MAX98 360A	Fit	0	0	2	2	3	3	MAX98 360A s mall-sig nal/idle audio all ocation
U21	C024b	TPS229 17DBV R	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
LS1	C025	8 ohm 0.5-1 W speaker	TBD	0	0	13	13	27	27	single speaker acoustic -output average
LS2	C026	2nd speaker pad	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
U22	C027	DRV260 5L	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
M1	C028	LRA/ER M motor	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
BT1	C029	LP4511 65 300mAh (R)	HOLD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
BT2	C030	LP4511 65 300mAh (L)	HOLD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
F1	C031a	PTC/fus e (R branch)	TBD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
F2	C031b	PTC/fuse (L branch)	TBD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS2	C044-RCL	10 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS3	C044-RCR	10 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS1	C044-RBAT	10 mohm 1% 1W	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R21	C032-BYP	0R (pack-PCM baseline)	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U23	C032	BQ2970	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
Q1	C032-Q	Dual N-MOSFET (b2b)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R22	C032-Rv	330R	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
C2	C032-Cv	0.1uF	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R23	C032-Rm	2k	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
RT1	C033a	10k NTC (R cell)	Fit	0.5	0.5	0.5	0.5	0.5	0.5	fuel-gauge/NTC/ protection monitor allocation
RT2	C033b	10k NTC (L cell)	Fit	0.5	0.5	0.5	0.5	0.5	0.5	fuel-gauge/NTC/ protection monitor allocation
TP2	C033-T P	Battery test points	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
J1	C034	Magnetic pogo 4-6p	TBD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
J2	C035	USB-C 16p USB2-only	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U24	C036	TPD2E001	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R1	C037a	5.1k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R2	C037b	5.1k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
J3	C038	Hirose FH26W 0.3mm FPC	HOLD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
J4	C039	Custom hinge FPC 6-10mm	HOLD	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
J5	C040	U.FL / I-PEX MHF	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
TP1	C041	UART/SWD pogo pads	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
SW1	C042a	Power/PWRKEY	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
SW2	C042b	Reset	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
SW3	C042c	MaskROM/Recovery	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
RS4	C044-R SOC	10 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS5	C044-R WIFI	50 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS6	C044-R AUD	50 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS7	C044-R C1	100 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
RS8	C044-R C8	100 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
RS9	C044-R C29	100 mohm 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U25	C044a	INA238 (I_BAT_TOTAL)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U26	C044-IC L	INA238 (I_CELL_L)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U27	C044-IC R	INA238 (I_CELL_R)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U28	C044c	INA238 (I_SOC_5V)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_off	quick_standby	phone_assist	mixed_active	record	ai_record	Basis
U29	C044e	INA238 (I_WIFI)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U30	C044f	INA238 (I_AUDIO)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U31	C044-IC 1	INA238 (I_CAM_1V1)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U32	C044-IC 8	INA238 (I_CAM_1V8)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
U33	C044-IC 29	INA238 (I_CAM_2V9)	DNP	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
R11	C046a	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R12	C046b	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R13	C046c	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R14	C046d	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R15	C046e	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
R16	C046f	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R17	C046g	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R18	C046h	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R19	C046i	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R20	C046j	100k	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
R3	C045a	2.2k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R4	C045b	2.2k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R5	C045c	2.2k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R6	C045d	2.2k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
R9	C045e	2.2k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain

Ref	BOM ID	Value	Status	deep_o ff	quick_s tandby	phone_ assist	mixed_ active	record	ai_reco rd	Basis
R10	C045f	2.2k 1%	Fit	0	0	0	0	0	0	0 direct load, DNP, passive, source, or included in owning domain
TOTAL				**22**	**134**	**362**	**450**	**1290**	**2065**	Current model total

8. Power domains and shutdown policy

Domain	Rails / source	Main loads	Default state	Owner / control
Battery / pack	LP451165 1S2P -> F1/F2 -> RS2/RS3 -> BAT_P -> RS1 -> NPM_VBAT	Entire system source	Live	Pack protection + nPM1300
AON	AON_1V8, AON_3V3, AON_LSW2 from nPM1300	nRF54L15, NDP120, BMI270, wake mic, fuel/NTC	On in normal standby	nPM1300 / nRF54L15
RK3576 compute	VSYS -> RS4 -> TPS61088 -> SOC_5V -> RK806S rails	RK3576, LPDDR4X, eMMC	Off until `SOC_PWR_EN` + `PMIC_PWRON`	nRF54L15 master; Linux safe-off handshake
Camera	TPS62840 CAM_1V1, TPS22917 CAM_1V8, TLV75529 CAM_2V9	IMX415 custom FPC module	Off until camera session	nRF54L15 / RK3576
Wi-Fi	VSYS -> RS5 -> TPS62825 -> TPS22917 -> FCU760K	FCU760K and antenna path	Off until transfer/preview	nRF54L15 enables buck, switch, and module
Audio	VSYS -> RS6 -> TPS22917 -> MAX98360A -> LS1	Single 8 ohm speaker	Off/muted	nRF54L15 / RK3576

Default-off controls use 100k pull-downs R11-R20 so compute, camera, Wi-Fi, and audio remain off if firmware is absent or reset.

9. Layout-entry gates

Gate	Topic	Current status	Close condition
G00	Mechanical/battery	HOLD	Real cell fit, swell room, FPC corridor, antenna keep-out, no battery over RK3576/PMIC
G00F	Phase 1.5 floorplan	HOLD	Dimensioned top/side floorplan + no-route KiCad placement envelopes prove fit before routing
G01	RK3576 identity + HDG	OPEN	Full datasheet, HDG, ball map, package drawing, reference delta
G02	LPDDR4X	OPEN	MPN, topology, placement, length report, DDR review
G03	RK806S PMIC	OPEN	Exact MPN, rails, inductors/caps, sequencing, timing
G04	eMMC/boot	OPEN	MPN, bootloader/BSP support, cold boot and power-loss recovery
G05	Wi-Fi FCU760K	HOLD	Hardware design, land pattern/3D, antenna SKU, BSP driver/firmware enumeration
G06	High-power transient sizing	OPEN	RK3576 boot/AI peak, boost soft-start, 1S droop, Wi-Fi/camera/audio peaks
G07	LP451165 + 1S2P pack	HOLD	Formal datasheet, discharge curves, IR/cycle/cert, pairing/f using/NTC/current-share
G08	AON power	HOLD	nPM1300 EK config, NDP120 kit, measured AON ≤ 25 mW avg / ≤ 50 mW ceiling
G09	Thermal/boost droop	OPEN	RK3576 burst thermal path, TPS61088/RK806 droop and UVLO margin
G10	Camera module	HOLD	Final IMX415 module FPC pinout, lens/FOV, lane count, rail current/timing
G11	Mic/audio topology	OPEN	Mic coordinates, ports, wind/AEC/beamforming, NDP120 wake path
G12	FPC/interconnect	HOLD	Pin count, impedance, bend radius/life if foldable version is required

Gate	Topic	Current status	Close condition
G13	Magnetics/passives height	OPEN	Inductor Isat/DCR/height, capacitor derating, wearable Z limits
G14	RF/worn tune	OPEN	Antenna SKU, keep-out, matching in shell and worn condition
G15	Passive/manufacturing BOM	OPEN	Expanded R/C/L MPNs, derating, lifecycle and alternates

10. Detailed schematic cards

U1 / C001 - RK3576

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:VERIFY_FCCSP698L_16x17
- Gate: G01 (RK3576 datasheet/HDG/ballmap/DDR guide)
- Description: Main SoC - 8-core, 6 TOPS NPU, ISP, MIPI CSI, H.264/265. Demand-started compute island (NOT always-on). Functional block: interface pins only.
- Note: FCCSP698L 16.1x17.2mm 0.6mm. Ball map not yet released -> only architecturally-defined interface pins are drawn. Start as a delta off the Radxa reference (Phase 2).
- Schematic nets: VDD_CPU->VDD_CPU, VDD_GPU->VDD_GPU, VDD_NPU->VDD_NPU, VDD_LOGIC->VDD_LOGIC, VDD_DDR->VDD_DDR, VCC_DDRIO->VCC_DDRIO, VCCIO_1V8->VCCIO_1V8, VCC_3V3->VCC_3V3, GND->GND, XIN->SOC_XIN, XOUT->SOC_XOUT, PMIC_SLEEP->PMIC_SLEEP, SHUTDOWN_REQ->SOC_SHUTDOWN_REQ, SAFE_TO_OFF->SOC_SAFE_TO_OFF, ALIVE->SOC_ALIVE, FAULT->SOC_FAULT, AON_UART_RX->AON_UART_TX, AON_UART_TX->AON_UART_RX, DDR_BUS->DDR_BUS, EMMC_BUS->EMMC_BUS, FSPI_BUS->FSPI_BUS, CSI_CLK_P->CSI_CLK_P, CSI_CLK_N->CSI_CLK_N, CSI_D0_P->CSI_D0_P, CSI_D0_N->CSI_D0_N, CSI_D1_P->CSI_D1_P, CSI_D1_N->CSI_D1_N, CAM_I2C_SCL->CAM_I2C_SCL, CAM_I2C_SDA->CAM_I2C_SDA, CAM_RST_L->CAM_RST_L, CAM_PWDN_L->CAM_PWDN_L, CAM_MCLK->CAM_MCLK, PDM_ARR_CLK->PDM_ARRAY_CLK, PDM_ARR_D0->PDM_ARRAY_D0, PDM_ARR_D1->PDM_ARRAY_D1, I2S_BCLK->I2S_BCLK, I2S_LRCLK->I2S_LRCLK, I2S_DIN->I2S_DIN, SARADC_NTC->SOC_NTC, USB2_WIFI_DP->WIFI_USB_DP, USB2_WIFI_DM->WIFI_USB_DM, PCM_CLK->WIFI_PCM_CLK, PCM_SYNC->WIFI_PCM_SYNC, PCM_DIN->WIFI_PCM_DIN, PCM_DOUT->WIFI_PCM_DOUT, WIFI_WAKE_L->WIFI_WAKE_L, USB2_DP->USB2_DP, USB2_DM->USB2_DM, DBG_UART_TX->SOC_DBG_TX, DBG_UART_RX->SOC_DBG_RX, MASKROM_n->SOC_MASKROM_L, RESET_n->SOC_RESET_L, PWRKEY->SOC_PWRKEY

U2 / C002 - RK806S-5

- Board/status: R-Temple Compute Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_QFN

- Gate: G03 (RK806S MPN/inductors/caps/timing per Radxa/Rockchip FAE)
- Description: SoC PMIC - generates CPU/GPU/NPU/DDR/IO rails + power sequencing for RK3576. Reuse verified RK3576 power architecture; NOT for AON.
- Note: Input SOC_5V from TPS61088. Rail names are the Radxa RK806S reference set; exact rail count/voltages freeze at G02/G03.
- Schematic nets: VIN->SOC_5V, GND->GND, PWRON->PMIC_PWRON, SLEEP->PMIC_SLEEP, I2C_SCL->SOC_PMIC_SCL, I2C_SDA->SOC_PMIC_SDA, RESET_n->SOC_RESET_L, VDD_CPU->VDD_CPU, VDD_GPU->VDD_GPU, VDD_NPU->VDD_NPU, VDD_LOGIC->VDD_LOGIC, VDD_DDR->VDD_DDR, VCC_DDRIO->VCC_DDRIO, VCCIO_1V8->VCCIO_1V8, VCC_3V3->VCC_3V3

U3 / C003 - LPDDR4X 4GB (MPN TBD)

- Board/status: R-Temple Compute Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_BGA
- Gate: G02 (LPDDR4X MPN/topology/placement, DDR review + length report)
- Description: System memory - 4 GB LPDDR4X x32, placed adjacent to RK3576. Reuse a Rockchip/Radxa-verified MPN to lower first chip-down risk vs LPDDR5.
- Schematic nets: VDD_DDR->VDD_DDR, VDDQ->VCC_DDRIO, GND->GND, DDR_BUS->DDR_BUS

U4 / C004 - eMMC 5.1 32GB (MPN TBD)

- Board/status: R-Temple Compute Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_BGA153
- Gate: G04 (eMMC MPN + BSP/bootloader; cold-boot + power-loss recovery)
- Description: System storage - 32 GB eMMC 5.1 (HS200/HS400 per BSP). Small + reliable; no microSD. Holds OS, models, event clips.
- Schematic nets: VCC->VCC_3V3, VCCQ->VCCIO_1V8, GND->GND, EMMC_BUS->EMMC_BUS

Y1 / C005 - 24 MHz 10 ppm XTAL

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:Crystal_SMD_3225
- Gate: G01 (reuse reference CL/ESR + placement distance per HDG)
- Description: Main clock - 24 MHz, 10 ppm; load per Rockchip reference layout.
- Schematic nets: XIN->SOC_XIN, XOUT->SOC_XOUT, GND->GND

U5 / C006 - MX25U6432F

- Board/status: R-Temple Compute Board / DNP / P1
- Package field: AI_Glasses_V2:USON-8
- Gate: Boot flow decides whether production needs it
- Description: Backup boot flash - 64 Mbit 1.8 V serial NOR on FSPI. Reserved for recovery / factory test / alt-boot; not populated on first EVT.

- Schematic nets: VCC->VCCIO_1V8, GND->GND, FSPI_BUS->FSPI_BUS

U6 / C007 - TPS61088

- Board/status: R-Temple Compute Board / HOLD / P0
- Package field: AI_Glasses_V2:VQFN-22_4.5x3.5
- Gate: G06/G09 (measured RK3576 boot+AI peak, soft-start, droop, thermal)
- Description: Main SoC 5 V boost - 1S -> SOC_5V for the compute island. High-current sync boost; EN from nRF54L15, PG back to nRF. Candidate/HOLD until peak measured.
- Note: Only the RK3576 island gets a boost (Phase-3 rule). Verify RK806 input range before committing the boost; if RK806 takes 1S directly, drop it.
- Schematic nets: VIN->SOC_IN, SW->SOC_BOOST_SW, GND->GND, EN->SOC_PWR_EN, PG->SOC_5V_PGOOD, VOUT->SOC_5V

L1 / C008 - Boost inductor (TBD)

- Board/status: R-Temple Compute Board / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_L_5x5
- Gate: G06/G13 (Isat covers boot+AI peak; height <=2mm; thermal)
- Description: Boost inductor for TPS61088 - low-DCR shielded, value per TPS61088 design. Do not freeze before transient sim/measurement.
- Schematic nets: A->SOC_IN, B->SOC_BOOST_SW

U11 / C015 - FCU760KAAMD

- Board/status: R-Temple Compute Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_LCC_13x12.2_Quectel
- Gate: G05 (Quectel HW Design + RK3576 BSP driver/firmware/enum) + official LCC land pattern
- Description: Wi-Fi 6 + BT 5.4 module - Quectel FCU760K, LCC 13.0x12.2x2.0mm. USB2 to RK3576 host; PCM for BT audio; VBAT 3.0-3.6V (typ 3.3V), max TX 353 mA. On-demand, load-switched off.
- Note: CONFIRMED on Radxa CM4 V1.20 p20; interface = USB2 (NOT SDIO). Single-antenna SKU FCU760KAAMD shares Wi-Fi/BT on ANT_WIFI_BT. Exact LCC pinout/land pattern + Linux driver/firmware stay HOLD pending Quectel Hardware Design (see pack 02_WiFi_FCU760K).
- Schematic nets: VBAT->WIFI_VBAT_3V3, GND->GND, USB_DP->WIFI_USB_DP, USB_DM->WIFI_USB_DM, CHIP_EN->WIFI_CHIP_EN, PCM_CLK->WIFI_PCM_CLK, PCM_SYNC->WIFI_PCM_SYNC, PCM_IN->WIFI_PCM_DOUT, PCM_OUT->WIFI_PCM_DIN, WAKE->WIFI_WAKE_L, ANT_WIFI_BT->WIFI_ANT

U12 / C016 - TPS62825

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:QFN_1.5x1.5
- Gate: G05/G06 (inductor/caps per module 353mA TX peak + ripple)
- Description: Wi-Fi dedicated 3.3 V buck - feeds FCU760K VBAT; >=500 mA capability for the 353 mA TX peak (pack note). EN by AON MCU. Avoids routing the radio through 5 V (Phase-3 rule).

- Schematic nets: VIN->WIFI_IN, SW->WIFI_BUCK_SW, GND->GND, EN->WIFI_BUCK_EN, VOUT->WIFI_3V3

L2 / C016c - Wi-Fi buck inductor (TBD)

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:VERIFY_L_2x2
- Gate: Per TPS62825 peak+ripple
- Description: Wi-Fi buck inductor - SW->3V3, value per TPS62825 design.
- Schematic nets: A->WIFI_BUCK_SW, B->WIFI_3V3

U13 / C016b - TPS22917DBVR

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: Package_TO_SOT_SMD:SOT-23-5
- Gate: G05 (turn-on sequence w/ module)
- Description: Wi-Fi load switch - gates WIFI_3V3 -> FCU760K VBAT with QOD + reverse block. EN by AON MCU.
- Schematic nets: VIN->WIFI_3V3, VOUT->WIFI_VBAT_3V3, EN->WIFI_LS_EN, GND->GND

C1 / C015-C - 47uF 6.3V X5R

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:C_0805
- Gate: Local bulk per Quectel HW Design
- Description: FCU760K VBAT bulk storage - 47 uF local reservoir next to the module (pack: 47uF + 1uF + 100nF). Rides the 353 mA TX burst.
- Schematic nets: A->WIFI_VBAT_3V3, B->GND

RT3 / C043 - 10k NTC B=3435

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Probe position per thermal sim
- Description: SoC/enclosure temp monitor - 10k NTC on SoC heat-spreader + skin side. Wearable must watch both die and skin temp.
- Schematic nets: A->SOC_NTC, B->GND

U7 / C009 - nRF54L15-QFN48

- Board/status: L-Temple AON/Power Board / Fit / P0
- Package field: AI_Glasses_V2:QFN-48_6x6
- Gate: EVT-frozen (RF layout/SDK/package to freeze early)

- Description: Low-power BLE MCU - ALWAYS ON. Owns the system state machine; controls RK3576/camera/Wi-Fi/audio power ENs; talks IMU/buttons/gauge; UART to RK3576, SPI to NDP120. Turns RK3576 from an always-on host into a demand accelerator.
- Note: nRF54L15 (not nRF52/53): 1.5MB NVM / 256KB RAM headroom, long-life platform, later WLCSP shrink. QFN48 for the debuggable EVT board.
- Schematic nets: VDD_1V8->AON_1V8, VDD_3V3->AON_3V3, GND->GND, ANT->BLE_ANT, SOC_PWR_EN->SOC_PWR_EN, SOC_5V_PGOOD->SOC_5V_PGOOD, PMIC_PWRON->PMIC_PWRON, CAM_1V1_EN->CAM_1V1_EN, CAM_1V8_EN->CAM_1V8_EN, CAM_2V9_EN->CAM_2V9_EN, WIFI_BUCK_EN->WIFI_BUCK_EN, WIFI_LS_EN->WIFI_LS_EN, WIFI_CHIP_EN->WIFI_CHIP_EN, AUDIO_LS_EN->AUDIO_LS_EN, AMP_SD->AUDIO_AMP_SD, SOC_SHDN_REQ->SOC_SHUTDOWN_REQ, SOC_SAFE_OFF->SOC_SAFE_TO_OFF, SOC_ALIVE->SOC_ALIVE, SOC_FAULT->SOC_FAULT, AON_UART_TX->AON_UART_TX, AON_UART_RX->AON_UART_RX, I2C_SCL->I2C_AON_SCL, I2C_SDA->I2C_AON_SDA, PMIC_INT_L->PMIC_INT_L, PMIC_SHPHLD->PMIC_SHPHLD, IMU_INT1->IMU_INT1, IMU_INT2->IMU_INT2, DSP_SCK->DSP_SPI_SCK, DSP_MOSI->DSP_SPI_MOSI, DSP_MISO->DSP_SPI_MISO, DSP_CS_n->DSP_SPI_CS_n, DSP_WAKE->DSP_WAKE, DSP_READY->DSP_READY, DSP_FAULT->DSP_FAULT, DSP_RST_n->DSP_RESET_L, HAPTIC_EN->HAPTIC_EN, SWDIO->NRF_SWDIO, SWDCLK->NRF_SWDCLK

U8 / C010 - nPM1300

- Board/status: L-Temple AON/Power Board / HOLD / P0
- Package field: AI_Glasses_V2:QFN_5x5
- Gate: G08 (AON <=25/50mW) - configure via nPM PowerUP on EK before board
- Description: AON PMIC - 1S charger + Power-Path + Fuel Gauge + dual Buck + LDO/Load-Switch + Ship/Hibernate. Powers the small AON world ONLY; NOT the RK3576 peak path.
- Note: VSYS is the power-path system node feeding the boost + gated bucks. Fuel gauge configured for the paralleled 600 mAh, not 300.
- Schematic nets: VBAT->NPM_VBAT, VBUS->USB_5V, VSYS->VSYS, GND->GND, BUCK1->AON_1V8, BUCK2->AON_3V3, LSW2->AON_LSW2, NTC_R->NTC_R, NTC_L->NTC_L, I2C_SCL->I2C_AON_SCL, I2C_SDA->I2C_AON_SDA, INT->PMIC_INT_L, SHPHLD->PMIC_SHPHLD

U9 / C011 - NDP120

- Board/status: L-Temple AON/Power Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_QFN_5x5
- Gate: Full datasheet + dev kit + measured listening power + NDA/licensing
- Description: Always-on Audio DSP - ultra-low-power wake-word + audio front-end; wakes the MCU over GPIO. Moves voice listening off RK3576. Keep bypass + DNP capability.
- Note: HOLD until hardware datasheet, rails/sequence, PDM/I2S detail, firmware/model tools, wake-word licensing, real listening power, MOQ/lead all in hand.
- Schematic nets: VDD_1V8->AON_1V8, VDD_3V3->AON_3V3, GND->GND, PDM_CLK->PDM_WAKE_CLK, PDM_DATA->PDM_WAKE_DATA, HOST_SCK->DSP_SPI_SCK, HOST_MOSI->DSP_SPI_MOSI, HOST_MISO->DSP_SPI_MISO, HOST_CS_n->DSP_SPI_CS_n, WAKE->DSP_WAKE, READY->DSP_READY, FAULT->DSP_FAULT, RESET_n->DSP_RESET_L

U10 / C012 - BMI270

- Board/status: L-Temple AON/Power Board / Fit / P0
- Package field: Package_LGA:LGA-14_2.5x3mm_P0.5mm
- Gate: Evaluate false-trigger under real frame vibration
- Description: 6-axis IMU - low-power motion detect; motion-interrupt wakes the MCU. Cycling/running state ID for wearables.
- Schematic nets: VDD->AON_1V8, VDDIO->AON_1V8, GND->GND, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, INT1->IMU_INT1, INT2->IMU_INT2

J6 / C018 - BLE 2.4 GHz FPC/PCB ant

- Board/status: Temple Rears (batt/spkr/ant) / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_ANT
- Gate: Worn-state tuning; clear of battery/metal
- Description: BLE antenna - 50 ohm, always-on. Physically separate from the Wi-Fi antenna.
- Schematic nets: ANT->BLE_ANT, GND->GND

J7 / C017 - Wi-Fi 2.4/5 GHz FPC ant

- Board/status: Temple Rears (batt/spkr/ant) / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_ANT
- Gate: Head-loaded RSSI/throughput/SAR; keep-out
- Description: Wi-Fi dual-band antenna - 50 ohm, worn-state tuned. Away from SoC/battery/speaker magnet/hinge.
- Schematic nets: ANT->WIFI_ANT, GND->GND

U14 / C019 - IMX415-AAQR module (custom FPC)

- Board/status: Front Sensor Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_CAM_MODULE
- Gate: G10 (module lens/FOV/FPC pinout/lane/supply/timing from vendor)
- Description: Main camera module - 8.46 MP, first target 1080p30/60, MIPI CSI. Custom small sensor+lens rigid island + FPC (NOT the dev module). Sensor rails (Sony datasheet): AVDD 2.9V 128/156mA, IOVDD 1.8V 3mA, DVDD 1.1V 187/250mA (~0.58W typ, ~0.77W max).
- Note: 2-lane CSI drawn for 1080p (sensor supports 4). Rails fed from per-rail Kelvin-sense nodes (_S). Custom-module FPC pinout/lane/module-power-tree freezes at G10; sensor chip currents are Sony datasheet, module adds LDO/clock/FPC losses (pack 04_IMX415).
- Schematic nets: DVDD_1V1->CAM_1V1_S, IOVDD_1V8->CAM_1V8_S, AVDD_2V9->CAM_2V9_S, GND->GND, CSI_CLK_P->CSI_CLK_P, CSI_CLK_N->CSI_CLK_N, CSI_D0_P->CSI_D0_P, CSI_D0_N->CSI_D0_N, CSI_D1_P->CSI_D1_P, CSI_D1_N->CSI_D1_N, SCL->CAM_I2C_SCL, SDA->CAM_I2C_SDA, XCLR_L->CAM_RST_L, PWDN_L->CAM_PWDN_L, INCK->CAM_MCLK

U15 / C020 - TPS62840

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:VERIFY_WSON
- Gate: Output current/noise per final module (DVDD ~250mA max + margin)
- Description: Camera DVDD 1.1 V buck - ultra-low-Iq (~60 nA) buck for the sensor core (DVDD 187/250mA), up to ~750 mA. EN by AON MCU / SoC. True power-down when camera off.
- Schematic nets: VIN->VSYS, SW->CAM_1V1_SW, GND->GND, EN->CAM_1V1_EN, VOUT->CAM_1V1

L3 / C020b - Cam 1V1 buck inductor (TBD)

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:VERIFY_L_1.6x0.8
- Gate: Per TPS62840 design
- Description: Camera 1.1 V buck inductor - SW->1V1.
- Schematic nets: A->CAM_1V1_SW, B->CAM_1V1

U16 / C021 - TLV75529PDRVR

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:WSON-6_2x2
- Gate: Verify vs IMX415 AVDD peak (156mA) + PSRR
- Description: Camera AVDD 2.9 V LDO - 500 mA low-noise analog supply (AVDD 128/156mA). Low-noise LDO, NOT a series resistor off 3.3V (pack section 23).
- Schematic nets: VIN->VSYS, EN->CAM_2V9_EN, GND->GND, VOUT->CAM_2V9

U17 / C022 - TPS22917DBVR

- Board/status: Front Sensor Board / Fit / P0
- Package field: Package_TO_SOT_SMD:SOT-23-5
- Gate: Check reverse block, ramp, QOD, logic level
- Description: Camera IOVDD 1.8 V load switch / isolation - low-leak switch from AON_1V8 for camera I/O (IOVDD ~3mA); Quick-Output-Discharge kills reverse feed when camera off (section 24).
- Schematic nets: VIN->AON_1V8, VOUT->CAM_1V8_SW, EN->CAM_1V8_EN, GND->GND

U18 / C023a - TPD4E05U06

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:USON
- Gate: Low-cap array near FPC entry
- Description: MIPI/FPC ESD (clk + D0) - ultra-low-cap 4-ch ESD at the FPC entry.
- Schematic nets: CLK_P->CSI_CLK_P, CLK_N->CSI_CLK_N, D0_P->CSI_D0_P, D0_N->CSI_D0_N, GND->GND

U19 / C023b - TPD4E05U06

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:USON
- Gate: Low-cap array near FPC entry
- Description: MIPI/FPC ESD (D1 + control) - ultra-low-cap ESD for the second lane pair.
- Schematic nets: D1_P->CSI_D1_P, D1_N->CSI_D1_N, RST->CAM_RST_L, MCLK->CAM_MCLK, GND->GND

MK1 / C013a - T5837 (wake mic)

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:LGA-5_3.5x2.65
- Gate: G11 (mic coords/ports/wind/wake + AEC/beamforming)
- Description: PDM wake mic - dedicated always-on wake channel into NDP120 (AON audio domain). Deep Off must capture without RK3576.
- Schematic nets: VDD->AON_3V3, GND->GND, CLK->PDM_WAKE_CLK, DATA->PDM_WAKE_DATA

MK2 / C013b - T5837 (array mic 1)

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:LGA-5_3.5x2.65
- Gate: G11
- Description: PDM array mic 1 - into RK3576 PDM for beamforming during vision/voice tasks.
- Schematic nets: VDD->AON_3V3, GND->GND, CLK->PDM_ARRAY_CLK, DATA->PDM_ARRAY_D0

MK3 / C013c - T5837 (array mic 2)

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:LGA-5_3.5x2.65
- Gate: G11
- Description: PDM array mic 2 - second array channel (L/R stereo on shared PDM clock).
- Schematic nets: VDD->AON_3V3, GND->GND, CLK->PDM_ARRAY_CLK, DATA->PDM_ARRAY_D1

MK4 / C014 - T5837 (4th mic)

- Board/status: Front Sensor Board / DNP / P1
- Package field: AI_Glasses_V2:LGA-5_3.5x2.65
- Gate: G11 (populate only if array sim/proto needs it)
- Description: 4th array mic - footprint only; on the switchable AON_LSW2 rail (nPM1300 load switch, section 14 'spare mic') so it can be fully powered down. Not populated until array/wind tests.
- Schematic nets: VDD->AON_LSW2, GND->GND, CLK->PDM_ARRAY_CLK, DATA->PDM_ARRAY_D1

U20 / C024 - MAX98360A

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:VERIFY_WLP_QFN
- Gate: Verify Z/cavity/EMI/peak power
- Description: Digital Class-D amp - I2S in, mono, low-power shutdown. Direct digital audio chain, no analog codec. Fully power-downable via load switch.
- Schematic nets: VDD->AUDIO_PWR, GND->GND, BCLK->I2S_BCLK, LRCLK->I2S_LRCLK, DIN->I2S_DIN, SD_MODE->AUDIO_AMP_SD, OUT_P->SPK_P, OUT_N->SPK_N

U21 / C024b - TPS22917DBVR

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: Package_TO_SOT_SMD:SOT-23-5
- Gate: Amp must be fully power-downable (V1 lesson)
- Description: Audio load switch - gates AUDIO_PWR to the amp; EN by AON MCU / SoC.
- Schematic nets: VIN->AUDIO_IN, VOUT->AUDIO_PWR, EN->AUDIO_LS_EN, GND->GND

LS1 / C025 - 8 ohm 0.5-1 W speaker

- Board/status: Temple Rears (batt/spkr/ant) / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_SPK
- Gate: MPN after acoustic cavity/volume/leak test
- Description: Main speaker - first version single speaker; open acoustic unit set by temple cavity.
- Schematic nets: P->SPK_P, N->SPK_N

LS2 / C026 - 2nd speaker pad

- Board/status: Temple Rears (batt/spkr/ant) / DNP / P1
- Package field: AI_Glasses_V2:VERIFY_SPK
- Gate: Decide dual-speaker at EVT-B
- Description: Second speaker land - reserved for productization; not populated first version.
- Schematic nets: P->NC, N->NC

U22 / C027 - DRV2605L

- Board/status: L-Temple AON/Power Board / DNP / P1
- Package field: Package_SO:MSOP-10_3x3mm_P0.5mm
- Gate: Vibration P1 - may DNP on first board (IMU/mic coupling)
- Description: Haptic driver - LRA/ERM driver, I2C, waveform library. Silent safety reminder path.
- Schematic nets: VDD->VSYS, GND->GND, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, EN->HAPTIC_EN, OUT_P->HAPTIC_P, OUT_N->HAPTIC_N

M1 / C028 - LRA/ERM motor

- Board/status: Temple Rears (batt/spkr/ant) / DNP / P1

- Package field: AI_Glasses_V2:VERIFY_LRA
- Gate: Vibration P1
- Description: LRA/ERM motor - short haptic alerts; can be replaced by phone vibration first version.
- Schematic nets: P->HAPTIC_P, N->HAPTIC_N

BT1 / C029 - LP451165 300mAh (R)

- Board/status: Temple Rears (batt/spkr/ant) / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_CELL_4.5x11x65
- Gate: G07 (full datasheet, >=2C, IR, cycles, cert)
- Description: Right cell - 3.7 V 300 mAh, 4.5x11x65 mm, high-rate custom. Right-temple weight.
- Note: One cell per temple, 1S2P = 600 mAh. Cell is a mechanical-size candidate; not a confirmed production battery until G07 datasheet + discharge curves land. Mechanical: use the 70x12.8x5.6 mm max clearance envelope, NOT the nominal 65x11x4.5 (pack 06).
- Schematic nets: +->BATR_P, -->CELL_NEG

BT2 / C030 - LP451165 300mAh (L)

- Board/status: Temple Rears (batt/spkr/ant) / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_CELL_70x12.8x5.6_maxenv
- Gate: G07
- Description: Left cell - same batch/capacity/IR-matched to the right cell. Left-temple weight. 1S2P parallel only after supplier pairing (OCV<=20mV, cap<=3%, DCIR<=10%, same lot, section 20).
- Schematic nets: +->BATL_P, -->CELL_NEG

F1 / C031a - PTC/fuse (R branch)

- Board/status: Temple Rears (batt/spkr/ant) / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_FUSE
- Gate: Rating from peak-current calc (> branch peak, < FPC rating)
- Description: Right branch protection - resettable PTC / fusible link in the right cell branch. Limits a single-cell / FPC short (section 7).
- Schematic nets: A->BATR_P, B->BATR_F

F2 / C031b - PTC/fuse (L branch)

- Board/status: Temple Rears (batt/spkr/ant) / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_FUSE
- Gate: Rating from peak-current calc
- Description: Left branch protection - resettable PTC / fusible link in the left cell branch.
- Schematic nets: A->BATL_P, B->BATL_F

RS2 / C044-RCL - 10 mohm 1%

- Board/status: Temple Rears (batt/spkr/ant) / Fit / P0
- Package field: AI_Glasses_V2:R_0805_shunt
- Gate: Branch current-share (section 7/section 29)
- Description: Left branch shunt - BATL_F -> BAT_P; I_CELL_L for current-share monitoring.
- Schematic nets: A->BATL_F, B->BAT_P

RS3 / C044-RCR - 10 mohm 1%

- Board/status: Temple Rears (batt/spkr/ant) / Fit / P0
- Package field: AI_Glasses_V2:R_0805_shunt
- Gate: Branch current-share (section 7/section 29)
- Description: Right branch shunt - BATR_F -> BAT_P; I_CELL_R for current-share monitoring.
- Schematic nets: A->BATR_F, B->BAT_P

RS1 / C044-RBAT - 10 mohm 1% 1W

- Board/status: L-Temple AON/Power Board / Fit / P0
- Package field: AI_Glasses_V2:R_1206_shunt
- Gate: I_BAT_TOTAL; the one production-kept sense path
- Description: Pack total shunt - BAT_P -> nPM1300 VBAT. Whole-device current (charge + discharge).
- Schematic nets: A->BAT_P, B->NPM_VBAT

R21 / C032-BYP - 0R (pack-PCM baseline)

- Board/status: L-Temple AON/Power Board / Fit / P0
- Package field: AI_Glasses_V2:R_0805
- Gate: Populate 0R OR discrete protection, never both (section 17)
- Description: Cell-negative bypass - ties CELL_NEG to pack GND when the supplier 1S2P pack already integrates its PCM. DNP this and populate U23+Q1 only if doing board-level protection.
- Schematic nets: A->CELL_NEG, B->GND

U23 / C032 - BQ2970

- Board/status: L-Temple AON/Power Board / DNP / P0
- Package field: Package_TO_SOT_SMD:SOT-23-5
- Gate: Keep ONE protection scheme; confirm suffix OVP/UVP/OC vs nPM1300/boost UVLO (section 17)
- Description: 1S protection IC (fallback) - OV/UV/OC/short via two low-side back-to-back NMOS. DNP if the supplier pack integrates protection (avoid double-protection conflict, section 17).
- Note: Wired per TI BQ2970 typical app (pack 03_Battery TI datasheet): VDD via Rvdd from BAT_P with 0.1uF to VSS(=CELL_NEG); V- via Rvm to pack GND; DO/CO drive Q1.
- Schematic nets: VDD->PROT_VDD, VSS->CELL_NEG, V-->PROT_VM, DO->PROT_DO, CO->PROT_CO

Q1 / C032-Q - Dual N-MOSFET (b2b)

- Board/status: L-Temple AON/Power Board / DNP / P0
- Package field: AI_Glasses_V2:VERIFY_DFN_2FET
- Gate: Vds/RDSon/Vgs/peak per BQ2970 app (section 17)
- Description: Back-to-back protection NMOS - discharge + charge FETs between CELL_NEG and pack GND; gates driven by BQ2970 DO/CO. DNP with U23.
- Schematic nets: D_CELL->CELL_NEG, G_DIS->PROT_DO, D_PACK->GND, G_CHG->PROT_CO

R22 / C032-Rv - 330R

- Board/status: L-Temple AON/Power Board / DNP / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Per BQ2970 app
- Description: BQ2970 VDD series R - from BAT_P (DNP with protection).
- Schematic nets: A->BAT_P, B->PROT_VDD

C2 / C032-Cv - 0.1uF

- Board/status: L-Temple AON/Power Board / DNP / P0
- Package field: AI_Glasses_V2:C_0402
- Gate: Per BQ2970 app
- Description: BQ2970 VDD-VSS decoupling (DNP with protection).
- Schematic nets: A->PROT_VDD, B->CELL_NEG

R23 / C032-Rm - 2k

- Board/status: L-Temple AON/Power Board / DNP / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Per BQ2970 app (V- sense)
- Description: BQ2970 V- sense R - to pack GND (DNP with protection).
- Schematic nets: A->GND, B->PROT_VM

RT1 / C033a - 10k NTC (R cell)

- Board/status: Temple Rears (batt/spkr/ant) / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: NTC curve matched to nPM1300 charger config
- Description: Right cell temperature - 10k NTC at the cell into nPM1300 NTC/AON ADC. Per-cell NTC (section 15).
- Schematic nets: A->NTC_R, B->GND

RT2 / C033b - 10k NTC (L cell)

- Board/status: Temple Rears (batt/spkr/ant) / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: NTC curve matched to nPM1300 charger config
- Description: Left cell temperature - 10k NTC at the cell into nPM1300 NTC/AON ADC.
- Schematic nets: A->NTC_L, B->GND

TP2 / C033-TP - Battery test points

- Board/status: Temple Rears (batt/spkr/ant) / Fit / P0
- Package field: AI_Glasses_V2:TestPad_Set
- Gate: Cell/pack voltage + NTC probe access (section 21)
- Description: Battery Kelvin test points - Cell-L+, Cell-R+, Pack+, Cell-neg, both NTCs for pairing / current-share / temperature verification.
- Schematic nets: CELL_L->BATL_P, CELL_R->BATR_P, PACK_P->BAT_P, CELL_NEG->CELL_NEG, NTC_L->NTC_L, NTC_R->NTC_R

J1 / C034 - Magnetic pogo 4-6p

- Board/status: Temple Rears (batt/spkr/ant) / TBD / P0
- Package field: AI_Glasses_V2:VERIFY_POGO
- Gate: Magnet direction, sweat corrosion, short, cycle life
- Description: Charge/data interface - magnetic pogo (5 V charge, optional USB2). Thinner + more waterproof than a temple USB-C.
- Schematic nets: VBUS->USB_5V, GND->GND, DP->USB2_DP, DM->USB2_DM

J2 / C035 - USB-C 16p USB2-only

- Board/status: EVT Debug Tail / Fit / P0
- Package field: AI_Glasses_V2:USB_C_16P_MidMount
- Gate: Connector height + shell opening co-freeze
- Description: EVT USB-C - 5 V sink + USB2 OTG (no USB3/DP). For MaskROM/ADB/firmware/lab supply.
- Schematic nets: VBUS->USB_5V, GND->GND, DP->USB2_DP, DM->USB2_DM, CC1->USB_CC1, CC2->USB_CC2

U24 / C036 - TPD2E001

- Board/status: EVT Debug Tail / Fit / P0
- Package field: Package_TO_SOT_SMD:SOT-23-3
- Gate: Keep ~90 ohm diff + continuous ref gnd
- Description: USB2 ESD - low-cap ESD on D+/D- near the connector.
- Schematic nets: DP->USB2_DP, DM->USB2_DM, GND->GND

R1 / C037a - 5.1k 1%

- Board/status: EVT Debug Tail / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: USB-C spec
- Description: USB-C CC1 Rd - 5.1k 1% sink pulldown. LOCK value + 1% (V1 audit rule).
- Schematic nets: A->USB_CC1, B->GND

R2 / C037b - 5.1k 1%

- Board/status: EVT Debug Tail / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: USB-C spec
- Description: USB-C CC2 Rd - 5.1k 1% sink pulldown.
- Schematic nets: A->USB_CC2, B->GND

J3 / C038 - Hirose FH26W 0.3mm FPC

- Board/status: Front Sensor Board / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_FPC_FH26W
- Gate: G12 (pin count from final cam lane/mic/power split)
- Description: Front<->temple high-speed FPC - carries MIPI CSI + PDM + I2C + control + camera power + GND across to the compute/AON boards. Documents the hinge-FPC pin list.
- Note: Pass-through connector: each pin taps the crossing net so the FPC pin count is explicit (G12). Two independent FPCs may split camera vs audio at layout.
- Schematic nets: CSI_CLK_P->CSI_CLK_P, CSI_CLK_N->CSI_CLK_N, CSI_D0_P->CSI_D0_P, CSI_D0_N->CSI_D0_N, CSI_D1_P->CSI_D1_P, CSI_D1_N->CSI_D1_N, CAM_I2C_SCL->CAM_I2C_SCL, CAM_I2C_SDA->CAM_I2C_SDA, CAM_RST_L->CAM_RST_L, CAM_PWDN_L->CAM_PWDN_L, CAM_MCLK->CAM_MCLK, PDM_WAKE_CLK->PDM_WAKE_CLK, PDM_WAKE_DATA->PDM_WAKE_DATA, PDM_ARR_CLK->PDM_ARRAY_CLK, PDM_ARR_D0->PDM_ARRAY_D0, PDM_ARR_D1->PDM_ARRAY_D1, VSYS->VSYS, AON_1V8->AON_1V8, AON_3V3->AON_3V3, GND->GND, CAM_1V1_EN->CAM_1V1_EN, CAM_1V8_EN->CAM_1V8_EN, CAM_2V9_EN->CAM_2V9_EN

J4 / C039 - Custom hinge FPC 6-10mm

- Board/status: Temple Rears (batt/spkr/ant) / HOLD / P0
- Package field: AI_Glasses_V2:VERIFY_HINGE_FPC
- Gate: G12 (impedance, bend radius, life, hinge interference)
- Description: Hinge FPC (L<->R) - carries battery power path, AON UART, control lines and GND between the two temples. Enables the split-and-fold structure.
- Note: Pass-through: taps the L<->R crossing nets so the hinge pin/power budget is explicit.
- Schematic nets: BAT_P->BAT_P, VSYS->VSYS, GND->GND, SOC_PWR_EN->SOC_PWR_EN, SOC_5V_PGOOD->SOC_5V_PGOOD, PMIC_PWRON->PMIC_PWRON, AON_UART_TX->AON_UART_TX, AON_UART_RX->AON_UART_RX,

SOC_SHDN_REQ->SOC_SHUTDOWN_REQ, SOC_SAFE_OFF->SOC_SAFE_TO_OFF,
 SOC_ALIVE->SOC_ALIVE, SOC_FAULT->SOC_FAULT, WIFI_BUCK_EN->WIFI_BUCK_EN,
 AUDIO_LS_EN->AUDIO_LS_EN

J5 / C040 - U.FL / I-PEX MHF

- Board/status: Temple Rears (batt/spkr/ant) / Fit / P0
- Package field: Connector_Coaxial:U.FL_Hirose_U.FL-R-SMT-1_Vertical
- Gate: Connector height + mating life
- Description: Micro RF connector - 50 ohm, Wi-Fi & BLE antenna debug/attach for EVT.
- Schematic nets: SIG->WIFI_ANT, GND->GND

TP1 / C041 - UART/SWD pogo pads

- Board/status: EVT Debug Tail / Fit / P0
- Package field: AI_Glasses_V2:TestPad_Set
- Gate: Voltage domain clearly labelled (no 3.3V into 1.8V IO)
- Description: UART/SWD test pads - RK3576 UART0 + nRF SWD + GND/VREF for bring-up (no connector height).
- Schematic nets: SOC_TX->SOC_DBG_TX, SOC_RX->SOC_DBG_RX, SWDIO->NRF_SWDIO, SWDCLK->NRF_SWDCLK, GND->GND

SW1 / C042a - Power/PWRKEY

- Board/status: EVT Debug Tail / Fit / P0
- Package field: Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2
- Gate: First chip-down must recover from boot failure
- Description: Power key - RK3576 PWRON via AON/PMIC.
- Schematic nets: A->SOC_PWRKEY, B->GND

SW2 / C042b - Reset

- Board/status: EVT Debug Tail / Fit / P0
- Package field: Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2
- Gate: Bring-up
- Description: Reset key - RK3576 RESET_n.
- Schematic nets: A->SOC_RESET_L, B->GND

SW3 / C042c - MaskROM/Recovery

- Board/status: EVT Debug Tail / Fit / P0
- Package field: Button_Switch_SMD:SW_Push_1P1T_NO_CK_KMR2
- Gate: Bring-up (recover from bad boot)
- Description: MaskROM/recovery key - force USB recovery boot.

- Schematic nets: A->SOC_MASKROM_L, B->GND

RS4 / C044-RSOC - 10 mohm 1%

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0805_shunt
- Gate: 5-20 mohm (2A x 10m=20mV; avoid boost UVLO)
- Description: RK3576 island shunt - VSYS -> SOC_IN (I_SOC_5V).
- Schematic nets: A->VSYS, B->SOC_IN

RS5 / C044-RWIFI - 50 mohm 1%

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0603_shunt
- Gate: 20-100 mohm
- Description: Wi-Fi island shunt - VSYS -> WIFI_IN (I_WIFI).
- Schematic nets: A->VSYS, B->WIFI_IN

RS6 / C044-RAUD - 50 mohm 1%

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0603_shunt
- Gate: 20-100 mohm
- Description: Audio island shunt - VSYS -> AUDIO_IN (I_AUDIO).
- Schematic nets: A->VSYS, B->AUDIO_IN

RS7 / C044-RC1 - 100 mohm 1%

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0603_shunt
- Gate: Per-rail Kelvin (section 23/section 29); DVDD ~250mA
- Description: Camera DVDD shunt - CAM_1V1 -> CAM_1V1_S (I_CAM_1V1).
- Schematic nets: A->CAM_1V1, B->CAM_1V1_S

RS8 / C044-RC8 - 100 mohm 1%

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402_shunt
- Gate: Per-rail Kelvin (section 23/section 29); IOVDD ~3mA
- Description: Camera IOVDD shunt - CAM_1V8_SW -> CAM_1V8_S (I_CAM_1V8).
- Schematic nets: A->CAM_1V8_SW, B->CAM_1V8_S

RS9 / C044-RC29 - 100 mohm 1%

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0603_shunt
- Gate: Per-rail Kelvin (section 23/section 29); AVDD ~156mA
- Description: Camera AVDD shunt - CAM_2V9 -> CAM_2V9_S (I_CAM_2V9).
- Schematic nets: A->CAM_2V9, B->CAM_2V9_S

U25 / C044a - INA238 (I_BAT_TOTAL)

- Board/status: L-Temple AON/Power Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A; production may keep this one
- Description: Whole-device current across RS1 (BAT_P->VBAT). Deep Off total = AON only, so it also backs the AON 20-50 mW check; nPM1300 fuel gauge cross-checks.
- Schematic nets: IN_P->BAT_P, IN_N->NPM_VBAT, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U26 / C044-ICL - INA238 (I_CELL_L)

- Board/status: Temple Rears (batt/spkr/ant) / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only (current-share)
- Description: Left branch current across RS2 - cell current-share vs SOC/temp.
- Schematic nets: IN_P->BATL_F, IN_N->BAT_P, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U27 / C044-ICR - INA238 (I_CELL_R)

- Board/status: Temple Rears (batt/spkr/ant) / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only (current-share)
- Description: Right branch current across RS3 - cell current-share vs SOC/temp.
- Schematic nets: IN_P->BATR_F, IN_N->BAT_P, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U28 / C044c - INA238 (I_SOC_5V)

- Board/status: R-Temple Compute Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only
- Description: RK3576 island monitor across RS4 - record/AI curves + wake energy.
- Schematic nets: IN_P->VSYN, IN_N->SOC_IN, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U29 / C044e - INA238 (I_WIFI)

- Board/status: R-Temple Compute Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only
- Description: Wi-Fi island monitor across RS5 - FCU760K TX avg vs 353mA peak.
- Schematic nets: IN_P->VSY, IN_N->WIFI_IN, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U30 / C044f - INA238 (I_AUDIO)

- Board/status: R-Temple Compute Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only
- Description: Audio island monitor across RS6 - speech avg vs music peak.
- Schematic nets: IN_P->VSY, IN_N->AUDIO_IN, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U31 / C044-IC1 - INA238 (I_CAM_1V1)

- Board/status: Front Sensor Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only
- Description: Camera DVDD monitor across RS7 - sensor core draw + off-leakage.
- Schematic nets: IN_P->CAM_1V1, IN_N->CAM_1V1_S, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U32 / C044-IC8 - INA238 (I_CAM_1V8)

- Board/status: Front Sensor Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only
- Description: Camera IOVDD monitor across RS8 - I/O draw + off-leakage.
- Schematic nets: IN_P->CAM_1V8_SW, IN_N->CAM_1V8_S, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

U33 / C044-IC29 - INA238 (I_CAM_2V9)

- Board/status: Front Sensor Board / DNP / P1
- Package field: AI_Glasses_V2:VSSOP-10
- Gate: Power Gate - EVT-A only
- Description: Camera AVDD monitor across RS9 - analog draw + off-leakage.
- Schematic nets: IN_P->CAM_2V9, IN_N->CAM_2V9_S, SCL->I2C_AON_SCL, SDA->I2C_AON_SDA, GND->GND

R11 / C046a - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: SOC_PWR_EN pull-down - RK3576 boost default OFF (main SoC off at boot).
- Schematic nets: A->SOC_PWR_EN, B->GND

R12 / C046b - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: PMIC_PWRON pull-down - RK806 stays off until AON MCU sequences it.
- Schematic nets: A->PMIC_PWRON, B->GND

R13 / C046c - 100k

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: CAM_1V1_EN pull-down - camera core buck default OFF.
- Schematic nets: A->CAM_1V1_EN, B->GND

R14 / C046d - 100k

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: CAM_1V8_EN pull-down - camera I/O load switch default OFF.
- Schematic nets: A->CAM_1V8_EN, B->GND

R15 / C046e - 100k

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: CAM_2V9_EN pull-down - camera analog LDO default OFF.
- Schematic nets: A->CAM_2V9_EN, B->GND

R16 / C046f - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402

- Gate: Default OFF (section 4)
- Description: WIFI_BUCK_EN pull-down - Wi-Fi buck default OFF.
- Schematic nets: A->WIFI_BUCK_EN, B->GND

R17 / C046g - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: WIFI_LS_EN pull-down - Wi-Fi load switch default OFF.
- Schematic nets: A->WIFI_LS_EN, B->GND

R18 / C046h - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4/section 9: FCU760K CHIP_EN default low)
- Description: WIFI_CHIP_EN pull-down - FCU760K held off until AON MCU enables it (pack section 9).
- Schematic nets: A->WIFI_CHIP_EN, B->GND

R19 / C046i - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: AUDIO_LS_EN pull-down - audio amp load switch default OFF.
- Schematic nets: A->AUDIO_LS_EN, B->GND

R20 / C046j - 100k

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Default OFF (section 4)
- Description: AUDIO_AMP_SD pull-down - MAX98360A shutdown default (amp muted at boot).
- Schematic nets: A->AUDIO_AMP_SD, B->GND

R3 / C045a - 2.2k 1%

- Board/status: L-Temple AON/Power Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Confirm vs total AON bus capacitance
- Description: AON I2C_SCL pull-up to AON_1V8 (nPM1300/BMI270/INA238 bus).
- Schematic nets: A->AON_1V8, B->I2C_AON_SCL

R4 / C045b - 2.2k 1%

- Board/status: L-Temple AON/Power Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Confirm vs total AON bus capacitance
- Description: AON I2C SDA pull-up to AON_1V8.
- Schematic nets: A->AON_1V8, B->I2C_AON_SDA

R5 / C045c - 2.2k 1%

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Confirm vs camera bus capacitance
- Description: Camera I2C SCL pull-up to CAM_1V8_SW.
- Schematic nets: A->CAM_1V8_SW, B->CAM_I2C_SCL

R6 / C045d - 2.2k 1%

- Board/status: Front Sensor Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Confirm vs camera bus capacitance
- Description: Camera I2C SDA pull-up to CAM_1V8_SW.
- Schematic nets: A->CAM_1V8_SW, B->CAM_I2C_SDA

R9 / C045e - 2.2k 1%

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Confirm vs SoC PMIC bus
- Description: SoC-PMIC I2C SCL pull-up to VCCIO_1V8 (RK3576<->RK806).
- Schematic nets: A->VCCIO_1V8, B->SOC_PMIC_SCL

R10 / C045f - 2.2k 1%

- Board/status: R-Temple Compute Board / Fit / P0
- Package field: AI_Glasses_V2:R_0402
- Gate: Confirm vs SoC PMIC bus
- Description: SoC-PMIC I2C SDA pull-up to VCCIO_1V8.
- Schematic nets: A->VCCIO_1V8, B->SOC_PMIC_SDA

Regeneration: `python3 v2_chipdown/scripts/generate_current_schematic_report.py` Then render with kidoc to PDF. Do not hand-edit the PDF; update the schematic master or YAML and regenerate.